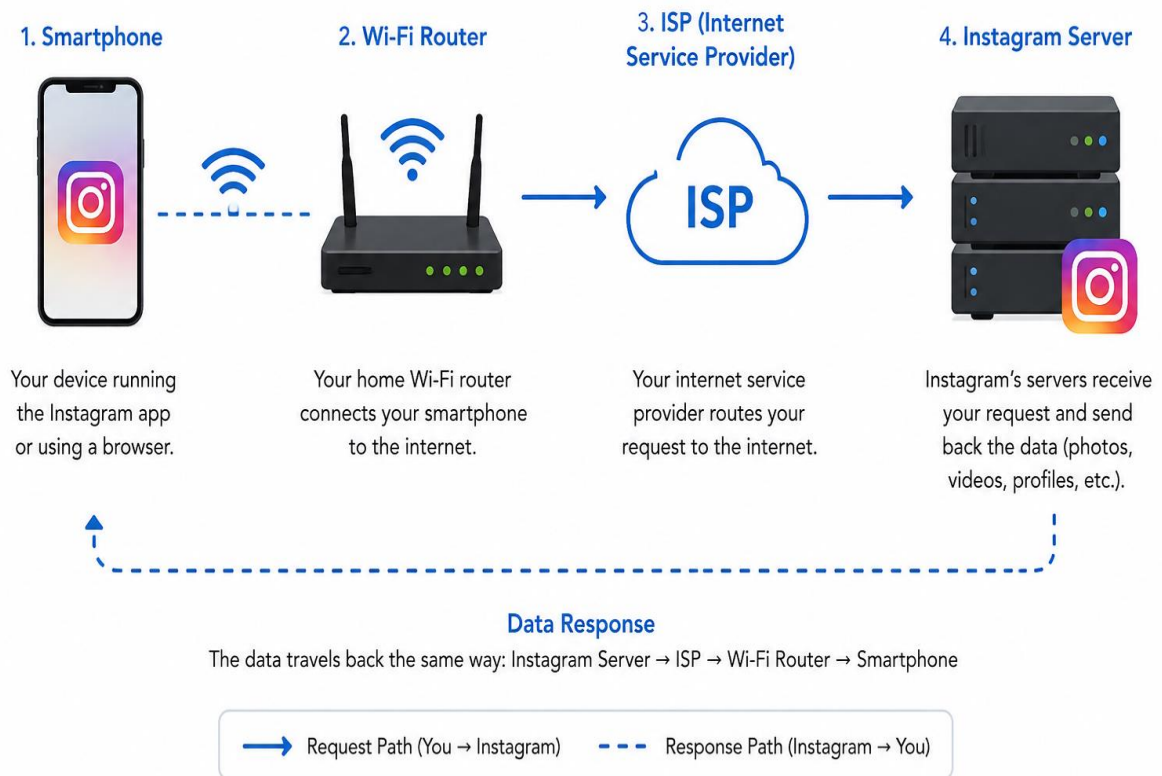


1. Draw a simple network diagram showing how your smartphone connects to Instagram — include your device, Wi-Fi router, ISP, and Instagram's server, and label each part.

How Your Smartphone Connects to Instagram



2. List the IP address of your device and your default gateway using the 'ipconfig' command (Windows) or 'ifconfig' (Linux/Mac), and explain what each means in 1-2 lines.
 - IPV4 Address – 192.168.1.20
 - This is the IP address assigned to my computer on the local network. It defines my device within that network.
 - Default Gateway – 192.168.1.1
 - This is usually router's IP address. A device sends traffic through the gateway when communicating outside the local network.

- 3. Simulate a Zomato food order journey by writing the steps/data packets that travel from your device to Zomato's server and back, mentioning protocols like HTTP, TCP/IP, and DNS wherever relevant.**

Zomato Food Order Journey

Example: You open the Zomato app, search for a restaurant, select food and place an order.

1) Device connects to the Internet

- Your phone/computer connects to the Wi-Fi router or mobile network.
- The router/ISP provides Internet connectivity.
- TCP/IP handles addressing and delivery of packets across networks.

2) DNS lookup

- Your device needs to find Zomato's server.
- It sends a DNS request asking, essentially "What IP address belongs to Zomato's server?"
- The DNS server returns the appropriate IP address.

3) Connecting to Zomato

- Your device establishes a connection with Zomato's server.
- TCP helps establish a reliable connection and ensures packet arrive correctly.
- The communication is normally protected using HTTPS.

4) Searching for a restaurant

- You type something like "Pizza near me".
- Your app sends an HTTPS request containing your search information.
- The request is broken into packets and travels as follows: Device/Wi-Fi router/ISP/Internet/Zomato Server

5) Zomato processes the request

- Zomato's server receives the packets.
- It processes the search and contains restaurant information from its database.
- The response is sent back as: Zomato Server/Internet/ISP/Router/Your Device.
- Your app displays the restaurants.

6) Selecting food

- You select a restaurant and add items to your cart.
- The app sends another HTTPS request to Zomato containing details such as restaurant ID, food-item IDs and cart information.

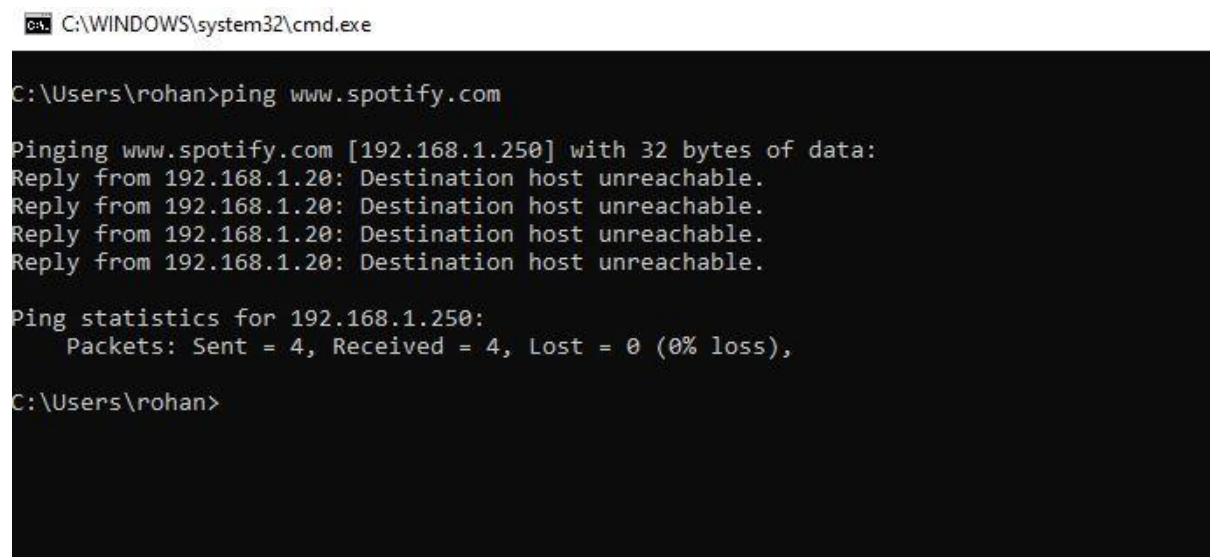
7) Placing the order

- When you tap “Place Order”. The app sends an HTTPS request containing the required order information.
- TCP/IP transports the packets reliably between the device and Zomato’s server.

8) Server confirms the order

- Zomato processes the order and sends a HTTPs response such as “Order Confirmed”.
- Your device receives the packets, reassembles them and the app displays the confirmation.

4. Use the 'ping' command to check the connection between your computer and www.spotify.com, and then explain what the results mean for network troubleshooting.



```
C:\WINDOWS\system32\cmd.exe

C:\Users\rohan>ping www.spotify.com

Pinging www.spotify.com [192.168.1.250] with 32 bytes of data:
Reply from 192.168.1.20: Destination host unreachable.
Reply from 192.168.1.20: Destination host unreachable.
Reply from 192.168.1.20: Destination host unreachable.
Reply from 192.168.1.20: Destination host unreachable.

Ping statistics for 192.168.1.250:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

C:\Users\rohan>
```

What the results mean

- DNS resolution- www.spotify.com was resolved to 192.168.1.250. This is private IP address, which is unusual for Spotify and may indicate a DNS/hosts-file entry or network configuration issue.

- Source computer- The message comes from 192.168.1.20 which appears to be computer's local IP address.
- Destination host unreachable- Your computer cannot reach 192.168.1.250.
- 4 packets received/0% loss- This is slightly misleading. The four packets received are "Destination host unreachable" error messages not successful replies from Spotify. Therefore, the actual connectivity test failed.
- No response time: There is no normal time=....ms response, confirming that the destination was not successfully reached.
- **Troubleshooting for Ping www.spotify.com**
- The test indicates a connectivity or DNS-resolution problem rather than a successful connection to Spotify. Here are the steps to troubleshoot this issue.
 - 1) Run ipconfig to check your IP address and default gateway.
 - 2) Run ping 192.168.1.1 to test connectivity to your router.
 - 3) Run ping 8.8.8.8 to test internet connectivity without relying on DNS.
 - 4) Run nslookup www.spotify.com to check which DNS server is resolving the Spotify address.
 - 5) Check the Windows hosts file for an incorrect Spotify entry.
 - 6) Check whether the router/DNS server is incorrectly resolving www.spotify.com to 192.168.1.250.